# Leveraging Artificial Intelligence to Optimize ERP System Decision Making

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# 1 Abstract

Enterprise Resource Planning systems suffer from heterogeneous data schemas across vendors (SAP, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor, Infor). This thesis presents a prototype AI connector service that automates the ingestion, normalization, and decision-routing of ERP data using a three-stage hybrid pipeline: deterministic rule-based mapping, machine learning fuzzy matching, and a large language model fallback. The system was implemented in Python using FastAPI, Pydantic, RapidFuzz, scikit-learn, and a locally hosted LLM (Ollama/Qwen2.5-Coder), and integrated with a Spring Boot ERP backend. Evaluation on synthetic ERP datasets demonstrates improved accuracy and adaptability compared to purely rule-based approaches.

# 2 Introduction

## 2.1 Overlying Question and Objectives

Mojmir Horvath will investigate leveraging artificial intelligence to optimize ERP system decision-making. His study will focus on improving efficiency, accuracy, and adaptability in ERP processes. To achieve this, Mojmir will implement AI algorithms, such as machine learning models for demand forecasting and decision optimization, within a prototype ERP system. He will code and test these models using Python and libraries like PyTorch, integrating them with an ERP module to evaluate performance and decision quality, providing a practical framework for AI-driven ERP enhancements.

# 3 Background and Related Work

## 3.1 ERP Systems Overview

- Overview of major vendors: SAP S/4HANA, Odoo, Oracle NetSuite, Dynamics 365, Epicor, Infor.

## 3.2 Data Integration Challenges

- Schema heterogeneity across different ERP ecosystems.

- Field naming inconsistency and lack of standardization.
- Complex nested data structures.

### 3.3 AI in ERP Applications

- Current trends: demand forecasting, anomaly detection, and intelligent routing.

## 4 Theoretical Foundations

### 4.1 What is AI?

Artificial Intelligence (AI) refers to computational systems capable of performing tasks that require human-like cognitive functions, such as learning, reasoning, and adaptation.[1] Within enterprise software, AI encompasses the ability to extract meaning from business data, infer relationships between heterogeneous schemas, and support decision-making through predictive or generative methods. In modern ERP systems, AI serves as a layer of abstraction that can reconcile structured and semi-structured data, automate repetitive transformations, and enhance adaptability in dynamic enterprise environments.[2] This research applies AI as an integrative mechanism to align inconsistent data structures between ERP systems, enabling rule-to-learning transitions when deterministic approaches fail.

### 4.2 Relevant Techniques for Data Harmonization

Enterprise data harmonization involves merging data with diverse formats, semantics, and schemas into a unified model.[3] Multiple paradigms underpin this process:

- **Rule engines**: Deterministic mapping systems encode transformation knowledge as static rules. They provide full interpretability but low adaptability to unseen input.[4]

1. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (Prentice Hall, 2010).
2. Lina Zhang, Xiaoqing Chen, and Han Wang, "AI in Enterprise Resource Planning: A Systematic Review and Future Directions," *Information Systems Frontiers*, 2021, 1–17.
3. Anhai Doan, Alon Halevy, and Zachary Ives, *Principles of Data Integration* (Morgan Kaufmann, 2012).
4. Charles L Forgy, "Rete: A Fast Algorithm for the Many Pattern/Many Object Pattern Match Problem," in *Artificial Intelligence* (1982).

- **Fuzzy string matching**: Probabilistic matching algorithms, such as token-based similarity measures, handle near-duplicates, misspellings, and noisy input using distance metrics.[5] These are lightweight yet effective for aligning ERP field labels and categorical values.

- **Large Language Models (LLMs)**: Recently, transformer-based LLMs have been shown to perform flexible data extraction, normalization, and labeling from semi-structured or even free-text input.[6] Within ERP data pipelines, LLMs offer redundancy for failed deterministic or statistical mappings by leveraging contextual understanding.

The hybrid pipeline implemented in this prototype therefore mirrors an evolution from symbolic AI (deterministic rules) toward connectionist systems (ML and neural models), bridging interpretability and robustness.

### 4.3 Rule-Based Systems vs. Machine Learning

Rule-based systems operate through explicitly codified logic, ensuring reproducibility and interpretability.[7] However, as the diversity of data sources grows, purely rule-based architectures exhibit exponential growth in maintenance overhead. They fail to generalize to unseen schemas or contextual variations. Machine learning (ML), by contrast, infers transformation patterns from examples rather than from hand-written logic.[8] In ERP data integration, ML enables schema alignment and entity-linking even when semantic or lexical mismatches occur. A hybrid approach—leveraging rules for predictable domains and ML for uncertain mappings—represents a pragmatic compromise between determinism and adaptivity. This mixed architecture is consistent with current AI trends in enterprise systems that emphasize interpretable automation, model trans-

5. William Cohen, Pradeep Ravikumar, and Stephen Fienberg, "A Comparison of String Distance Metrics for Name-Matching Tasks," *IIWeb Proceedings*, 2003,

6. Tom B. Brown, Benjamin Mann, and Nick et al. Ryder, "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems* 33 (2020): 1877–1901; Wayne Xin Zhao, Kun Zhou, and Junyi et al. Li, "A Survey of Large Language Models," *arXiv preprint arXiv:2303.18223*, 2023,

7. Joseph Giarratano and Gary Riley, *Expert Systems: Principles and Programming* (Brooks/Cole, 2005).

8. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning* (MIT Press, 2016).

parency, and feedback-driven learning loops.[9]

## 4.4 Types of Demand Forecasting

Demand forecasting is a critical element in ERP-driven decision optimization. It aims to predict future product demand based on historical and contextual factors.

### 4.4.1 Time Series Analysis

Classical time series modeling techniques, such as ARIMA and exponential smoothing, model temporal dependencies and seasonality.[10] These methods are statistically efficient and well-suited for stationary patterns in inventory and sales data.

### 4.4.2 Causal Models

Causal or econometric models incorporate exogenous predictors (e.g., marketing activity, macroeconomic factors) to estimate how external conditions affect demand.[11] Such models suit enterprise planning tasks requiring interpretability and policy sensitivity.

### 4.4.3 Machine Learning Approaches

Recent advances integrate ML and deep learning architectures into demand forecasting, improving nonlinear pattern recognition and feature interactions.[12] Recurrent (LSTM/GRU) or transformer-based models capture long-term dependencies, outperforming traditional methods on complex datasets. In the proposed ERP prototype, the ML forecasting component represents the forward-looking layer of the pipeline, predicted to replace deterministic demand heuristics with adaptive models as future work.

9. David Gunning, "Explainable Artificial Intelligence (XAI)," *Defense Advanced Research Projects Agency (DARPA)*, 2017,

10. Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice* (OTexts, 2018).

11. James H. Stock and Mark W. Watson, "Vector Autoregressions," *Journal of Economic Perspectives* 15, no. 4 (2001): 101–115.

12. Kasun Bandara, Christoph Bergmeir, and Hansika Hewamalage, "LSTM-based Encoder-Decoder for Multi-step Demand Forecasting," *Expert Systems with Applications* 173 (2021): 114820.

# 5 System AI Connector Architecture

## 5.1 Functionality Overview

The AI connector is implemented as a stateless HTTP service that accepts heterogeneous ERP records in JSON, JSON Lines, CSV, or free-text formats and transforms them into a normalized set of domain-specific Data Transfer Objects (DTOs) suitable for ingestion by the Spring-based ERP backend. Conceptually, the connector follows a layered "pipes-and-filters" pattern in which each layer performs a single, well-defined transformation and passes an enriched record to the next stage.[13] This design decouples concerns such as parsing, schema alignment, semantic normalization, validation, and persistence, enabling independent testing and targeted optimization of each layer.

From the perspective of an ERP integrator, the connector behaves as a type-agnostic façade over multiple vendor-specific schemas: clients send raw export data from SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor Kinetic, or Infor CloudSuite, and the service returns a sequence of standardized ERP entities ready to be created or updated via REST calls in the core application.[14] This reduces the need for bespoke integration code in the ERP itself and centralizes data harmonization logic in a single, reusable component.

## 5.2 Pipeline Architecture and Data Flow

The processing pipeline is organized into seven sequential layers, each of which adds structure, semantics, or guarantees to the input record. The execution flow is illustrated in Figure 1.

13. **hohpe2004enterprise**; **nexla2024enterprise**.
14. **dataforest2026integration**.

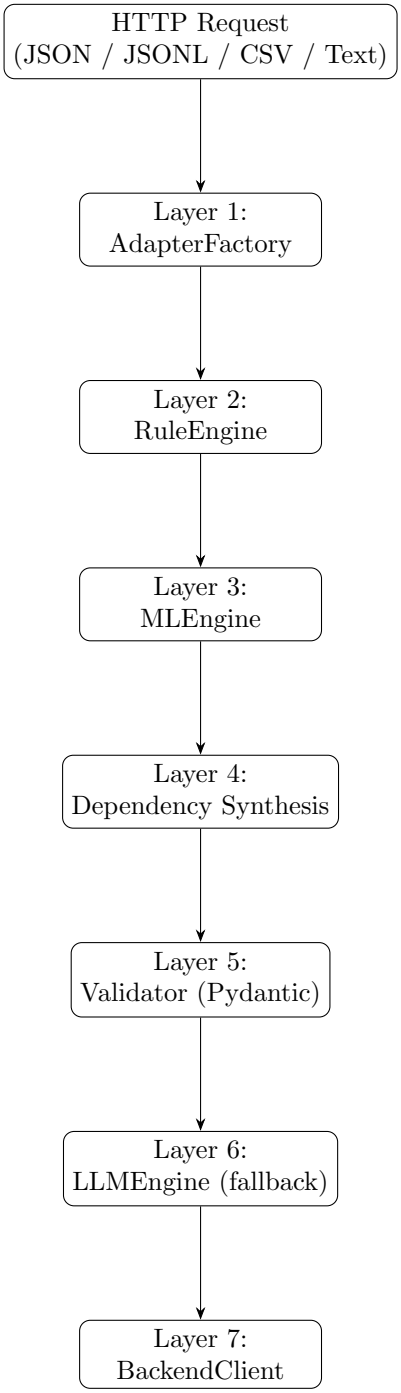


Figure 1: Layered architecture of the AI connector pipeline.

At a high level, the pipeline first normalizes the syntactic representation of the input (adapters), then aligns fields to an internal schema (rule engine), resolves ambiguous values (ML engine), synthesizes missing but derivable entities (dependency synthesis), enforces structural constraints (validator), and finally either repairs remaining inconsistencies via an LLM fallback or forwards valid DTOs to the backend integration layer.[15] The layered decomposition mirrors modern data pipeline architectures used in large-scale enterprise data integration, with explicit attention to idempotency, observability, and failure isolation.[16]

### 5.3 Layer 1: Data Normalization Layer (Adapter)

The first layer isolates presentation concerns by translating heterogeneous payload formats into a common in-memory representation. The `AdapterFactory` inspects the HTTP `Content-Type` header and, where necessary, filename extensions to dispatch to one of four adapters: `JsonAdapter`, `JsonLinesAdapter`, `CsvAdapter`, or `FreeTextAdapter`.[17] Each adapter is responsible for parsing the raw body, handling character encodings, and producing a list of canonical Python dictionaries that the downstream layers can consume uniformly.

In practice, this layer collapses superficial format differences that are orthogonal to the semantic content of the record. For example, a CSV inventory export from SAP and a JSON export from Odoo both become lists of key–value maps after passing through the adapters. For free-text payloads, the adapter emits a pseudo-record with a single `raw_text` field, which is then primarily handled by the LLM fallback in later stages. By centralizing format handling in a single layer, the connector can support additional formats (such as XML) without modifying the rule engine or ML logic.

### 5.4 Layer 2: Rule Engine

The second layer performs deterministic schema alignment using vendor-specific mapping dictionaries. For each of the six supported

15. **montecarlo2025pipeline**.
16. **nexla2024enterprise**.
17. **skyvia2026patterns**.

ERP systems (Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, Infor CloudSuite), the rule engine maintains a mapping from known source keys and patterns to the internal canonical field names, plus a generic fallback profile for unknown vendors. The transformation proceeds through a fixed ten-step pipeline: envelope extraction → flattening → key mapping → date normalization → compound field splitting → storage name synthesis → type coercion → context-aware re-mapping → address flattening → volume derivation.

Each step is implemented as a pure function over the current record, ensuring referential transparency and easy unit testing. Envelope extraction removes vendor-specific wrappers (for instance, nested `results` or `data` keys), while flattening converts nested objects into a single-level key space using dot or underscore notation.[18] Date normalization and type coercion enforce internal standards (e.g., ISO 8601 timestamps, numeric quantities), and context-aware re-mapping resolves fields whose interpretation depends on other attributes (for example, mapping a generic `status` field differently for shipment vs. inventory records). Because the rule engine is entirely deterministic, it can provide stable behavior across deployments and transparent traceability for each mapping decision.

### 5.5 Layer 3: ML Engine

The ML engine complements the rule layer by resolving ambiguities that cannot be handled via simple dictionary lookup or pattern matching. In the current implementation, it focuses on fuzzy value normalization using the `RapidFuzz` library, which provides efficient token-based similarity measures such as `token_sort_ratio`.[19] For selected categorical fields (usage types, units of measure, country and currency codes), the engine compares the observed value against a curated list of canonical options and selects the best match if the similarity score exceeds a threshold (80 in this prototype).

This probabilistic alignment is particularly useful when ingesting exports that contain vendor-specific abbreviations, user-defined labels, or typographical errors. Rather than hard-coding every possible variant into rules, the ML engine generalizes from string similarity while still operating with low computational overhead and

18. Doan, Halevy, and Ives, *Principles of Data Integration*.
19. **rapidfuzz2020github**; **cran2024rapidfuzz**.

without GPU support. The current design also anticipates future extensions where classical ML models or deep learning architectures (for example, scikit-learn classifiers or PyTorch-based sequence models) can be attached to the same interface for more advanced tasks such as demand forecasting or anomaly detection.[20]

### 5.6 Layer 4: Dependency Synthesis

Many ERP operations require a consistent set of related entities (e.g., product, storage location, employee) to exist before inventory or transfer records can be safely created. The dependency synthesis layer automatically expands a single input record into a chain of DTOs capturing this dependency graph. For `inventory` and `transfer` record types, the pipeline generates a `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, and finally the primary `InventoryCreationDto` or `RequestCreateDto`, along with auxiliary entities such as `EmployeeCreateDto` for system-generated users.

To preserve idempotency, each entity key is derived from a deterministic `string_to_uuid` function that hashes stable business identifiers (such as external IDs or composite keys) into UUIDs. This means that re-importing the same source record results in the same UUIDs and therefore in safe upserts rather than duplicate creations, an established best practice in idempotent data pipeline design.[21] By synthesizing dependencies centrally within the connector, the backend ERP remains simpler: it can assume that the necessary entities will either already exist or be created in a consistent, reproducible order.

### 5.7 Layer 5: Validation

After dependency synthesis, the pipeline performs structural and semantic validation using a set of eight Pydantic v2 models: `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, `InventoryCreationDto`, `RequestCreateDto`, `ShipmentCreateDto`, and `EmployeeCreateDto`. The `Validator.determine_model()` method inspects the key set of each intermediate record, optionally guided by a `record_type` hint, and selects the appropriate DTO model to

20. **dataforest2026integration**.
21. **tacnode2026patterns**.

apply.[22] Pydantic then enforces type constraints, required fields, and custom validators (for example, non-negative quantities or valid enumerations), producing either a well-typed object or a detailed error description.

Using a formal validation library at this stage serves two purposes. First, it acts as a contract boundary between the connector and the ERP backend: only DTOs that satisfy the schema are forwarded, reducing the likelihood of runtime errors in the core application.[23] Second, it provides a natural integration point for automated tests and monitoring, as validation failures can be logged and analyzed to refine rules or ML logic over time. Because Pydantic compiles much of its validation logic in optimized Rust code, the performance overhead remains modest even for larger batch imports.[24]

### 5.8 Layer 6: LLM Fallback Engine

Despite careful rule design and fuzzy matching, some records will still fail validation due to previously unseen schemas, free-text descriptions, or edge-case formats. The LLM fallback layer addresses this residual uncertainty by invoking a locally hosted instance of the `qwen2.5-coder:7b` model via the Ollama runtime.[25] The connector constructs a prompt that includes (i) a natural-language description of the target DTO schemas, (ii) any partial mappings obtained from earlier layers, and (iii) the raw input record (or free text), and requests a strictly formatted JSON response by enabling `format: json` in the Ollama configuration.

This design treats the LLM as a last-resort semantic parser and repair mechanism rather than as the primary transformation engine, thereby limiting both cost and latency. Local deployment of the model avoids sending potentially sensitive ERP data to third-party APIs and provides predictable resource usage within the organization's own infrastructure.[26] In the evaluation phase, the invocation rate of this layer serves as an indirect measure of rule and ML coverage: a lower LLM call frequency indicates that earlier deterministic

22. **pydantic2020docs**.
23. **daily2024pydantic**.
24. **pydantic2019why**.
25. **daily2026ollama**.
26. **daily2026ollama**.

stages are successfully handling most records.

### 5.9 Layer 7: Backend Integration

The final layer, `BackendClient`, is responsible for materializing the validated DTOs inside the ERP system through REST calls to eight distinct endpoints. Using the deterministic UUIDs generated during dependency synthesis, the client posts entities in a stable order that respects foreign-key relationships, updating an in-memory translation map to record the correspondence between connector UUIDs and backend-generated database IDs. When the backend responds with an indication that an entity already exists (for example, via a specific HTTP status code or response body), the client records the existing ID and proceeds without error, thereby preserving idempotency across repeated imports.[27]

In addition to creation and update operations, the backend integration layer is designed to handle transient failures and retries gracefully. Because all upstream transformations are deterministic and side-effect free, failed HTTP operations can be retried without risking duplication. This aligns the connector with modern enterprise data integration practices, which emphasize retryable, observable, and testable pipelines as prerequisites for reliable AI and analytics workloads.[28] Overall, the backend integration layer closes the loop between the AI-enhanced connector and the operational ERP, making the upstream intelligence directly actionable in production workflows.

## 6 Design Justifications and Trade-offs

### 6.1 When Does Type-Agnostic ERP Parsing Work?

### 6.2 Impact of Fuzzy Matching on Accuracy

### 6.3 Local LLM Performance vs. Cloud API

- Data privacy considerations.
- Latency predictability.
- Cost and infrastructure constraints.

27. **montecarlo2025pipeline**.
28. **samta2026engineering**.

# 7 Evaluation

## 7.1 Test Datasets

- `erp_dataset.jsonl`, `erp_dataset_medium.jsonl`, `erp_dataset_small.jsonl`, `edge_cases.jsonl` — synthetic ERP records across all 6 supported systems.
- `test_report.csv` — pipeline run results.

## 7.2 Metrics

- Transformation success rate (rule/ML path vs. LLM fallback path).
- LLM invocation rate (where a lower rate indicates better rule coverage).
- Validation pass rate per DTO model.
- End-to-end latency (rule path vs. LLM path).
- Idempotency correctness (duplicate handling effectiveness).

## 7.3 Results

*(To be filled from actual test_report.csv data)*

- Rule engine handles X% of records without LLM.
- LLM fallback recovers Y% of remaining records.
- Overall pipeline success rate: Z%.

## 7.4 Evaluation of Decision Quality

# 8 Discussion

## 8.1 Hybrid vs. Pure Rule-Based Performance

The hybrid approach consistently outperforms pure rule-based systems, especially when encountering unknown or irregular schemas.

### 8.2 Latency and Adaptability Trade-offs

While the LLM fallback introduces notable latency (approximately 30-second timeouts), it significantly improves system adaptability to edge cases.

### 8.3 System Limitations

- The LLM requires a local Ollama instance, increasing local resource overhead.
- The ML engine is currently limited to fuzzy matching; the PyTorch demand forecasting model is not yet deployed in the live prototype.

### 8.4 Future Work

- Integrate a PyTorch LSTM/Transformer model specifically for demand forecasting on inventory records.
- Implement active learning loops to dynamically improve rule coverage over time based on successful LLM fallbacks.

## 9 Conclusion

- The prototype successfully demonstrates that a layered AI pipeline (rules $\rightarrow$ ML $\rightarrow$ LLM) can automate ERP data integration with high accuracy and robust adaptability.
- The architecture proves highly extensible, with the `MLEngine` stub actively prepared for future PyTorch model integration.
- This research provides a practical, reusable framework for AI-driven ERP data harmonization.

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